CMAT · Formula Reference

Number System

Fractions

CONCEPT **Proper, Improper & Mixed**

Proper: Numerator smaller than denominator, value < 1 (e.g. 3⁄8).

Improper: Numerator ≥ denominator, value ≥ 1 (e.g. 9⁄4).

Mixed: Whole + proper fraction (e.g. 2 1⁄3). Convert by: (whole × denominator + numerator)⁄denominator → 2 1⁄3 = (2×3+1)/3 = 7⁄3.

FORMULA **Fraction → Percent**

$$\text{Percent} = \text{Fraction} \times 100\%$$

Multiplies the fraction by 100 to show “parts per 100”.

Example: 3⁄5 → 3⁄5 × 100 = 60 %. In the exam, always simplify first if possible to save time.

FORMULA **Adding Two Fractions**

$$\frac{a}{b} + \frac{c}{d} = \frac{a \times d + c \times b}{b \times d}$$

Here *a, c* are numerators and *b, d* are denominators. Use when denominators differ.

Example: 2⁄3 + 5⁄8 → (2×8 + 5×3)/(3×8) = (16 + 15)/24 = 31⁄24 = 1 7⁄24.

TRICK **Compare Fractions Instantly**

To decide which of a⁄b and c⁄d is bigger without calculating decimals, cross-multiply once:

If a × d > c × b, then a⁄b is larger; else c⁄d is larger.

Example: Which is bigger, 7⁄9 or 5⁄6? 7×6 = 42, 5×9 = 45 → 45 > 42, so 5⁄6 is larger.

FORMULA **Find a Fraction of a Quantity**

$$\text{Part} = \frac{m}{n} \times \text{Whole}$$

Gives the actual amount represented by m⁄n of a total.

Example: What is 3⁄8 of 200 kg? 3⁄8 × 200 = 600⁄8 = 75 kg.

EXAMPLE **Mixed → Improper Conversion**

Convert 4 2⁄5 to an improper fraction:

1. Whole part 4 × denominator 5 = 20
2. Add numerator: 20 + 2 = 22
3. Place over original denominator → 22⁄5.

Always useful before inserting into any algebraic equation.

Decimals

CONCEPT **Place Value after the Dot**

Immediately after the decimal:

- 1st place = Tenths (10^{-1})
- 2nd = Hundredths (10^{-2})
- 3rd = Thousandths (10^{-3})

Knowing this lets you switch between fractions and decimals in seconds.

FORMULA **Finite Decimal → Fraction**

$$\text{Fraction} = \frac{\text{Number without decimal point}}{10^{k}}$$

k = number of digits after the decimal.

Example: 0.375 → 375⁄1000. Simplify by 125 → 3⁄8.

FORMULA **Recurring Decimal → Fraction**

$$\text{Fraction} = \frac{N - P}{9\ldots90\ldots0}$$

N = all digits without the dot, P = non-repeating part. Denominator = as many 9s as repeating digits followed by as many 0s as non-repeating digits.

Example: $0.1\overline{6}$ (i.e. 0.1666…) → N = 16, P = 1, denominator = 90 → (16 − 1)/90 = 15/90 = 1/6.

TRICK **Quick × 10^n or ÷ 10^n**

Moving the decimal right by n places multiplies by 10^n. Moving left divides by 10^n.

Example: 3.47×100 = 347 (two places right), 58÷1000 = 0.058 (three places left).

FORMULA **Decimal → Percent**

$$\text{Percent} = \text{Decimal} \times 100\%$$

Same formula as for fractions: shift the decimal two places right and append %.

Example: 0.084 → 8.4 %.

EXAMPLE **Rounding Off**

Round 12.3487 to 2 decimal places:

1. Look at 3rd decimal (8). Since 8 ≥ 5, add 1 to 2nd place.
2. Result → 12.35.

Divisibility Rules

CONCEPT **Quick Checks for 2, 5 & 10**

- 2 → Last digit even.
- 5 → Last digit 0 or 5.
- 10 → Ends with 0.

Instant elimination in MCQs.

CONCEPT **Digit-Sum Rules (3 & 9)**

Add all digits.

- If sum divisible by 3 → number divisible by 3.
- If sum divisible by 9 → number divisible by 9.

Example: 7 2 6 → 7+2+6=15, divisible by 3 and 15≠multiple of 9, so 726 divisible by 3 but not by 9.

CONCEPT **Rules using Last Digits (4 & 8)**

4 → Last **two** digits form a number divisible by 4.

8 → Last **three** digits divisible by 8.

Example: 9 1 2 4: last two = 24 → yes for 4; last three = 124 → not divisible by 8.

CONCEPT **Rule for 6**

A number divisible by both 2 *and* 3 is divisible by 6. Check evenness and digit-sum in one glance.

TRICK **Fast Check for 7**

Double the last digit and subtract from the rest. Repeat if needed. If result is 0 or a multiple of 7, original number is too.

Example: 203 → 20 − (2×3)=14 → 14 divisible by 7 → 203 divisible by 7.

CONCEPT **Alternate-Sum Rule for 11**

Find |(sum of digits in odd places) − (sum in even places)|. If result is 0 or multiple of 11 → divisible by 11.

Example: 4 9 6 7 0 2 → (4+6+0)=10, (9+7+2)=18, |10−18|=8 (not multiple of 11) → not divisible.

FORMULA **Quick Remainder on 9**

$$\text{Remainder on } 9 = (\text{Digit sum}) \bmod 9$$

Add digits repeatedly until you get 0–8. That final single digit is the remainder when the original number is divided by 9.

Example: 8 5 7 2 → 8+5+7+2=22 → 2+2=4 → remainder = 4.

Factors

CONCEPT **What is a Factor?**

Whole numbers that divide the given number exactly (zero remainder). Factors come in pairs: if 6 divides 24, so does 24÷6 = 4.

FORMULA **Count of Factors from Prime Powers**

$$\text{Total factors of n} = (e_1 + 1)(e_2 + 1)(e_3 + 1)\dots$$

Prime-factorise n as $p_1^{e_1} \times p_2^{e_2} \times \dots$ Then add 1 to each exponent and multiply.

Example: $72 = 2^3 \times 3^2$ → (3+1)(2+1)=4×3=12 factors.

FORMULA **Sum of All Factors**

$$\text{Sum} = \frac{p_1^{e_1+1} - 1}{p_1 - 1} \times \frac{p_2^{e_2+1} - 1}{p_2 - 1} \dots$$

For each prime p with exponent e, build a mini-sum $(1+p+p^2+\dots+p^e)$ using the geometric-series shortcut above. Multiply all such sums.

Example: 72 again: $(2^4-1)/(2-1)=15$, $(3^3-1)/(3-1)=13$ → 15×13=195.

TRICK **Perfect Square Test via Exponents**

A number is a perfect square *iff* every prime exponent in its factorisation is even. Saves time versus taking square roots.

EXAMPLE **List All Factors of 36**

$36 = 2^2 \times 3^2$. Possible exponents: 0–2 for each prime.

Create pairs: $2^0 3^0 = 1$, $2^1 = 2$, $2^2 = 4$ with each 3-power 1,3,9 → factors: 1,2,3,4,6,9,12,18,36 (9 total). Cross-check with count formula: (2+1)(2+1)=9.

Multiples

CONCEPT **What is a Multiple?**

k is a multiple of m if k = m×integer. Multiples extend "outwards" by repeated addition of m (e.g. multiples of 4 → 4,8,12,16,…).

FORMULA

LCM Using HCF

$$\text{LCM}(a,b) = \frac{a \times b}{\text{HCF}(a,b)}$$

Works for two positive integers.

Example: For 12 and 18, HCF=6 → LCM=12×18/6=36.

FORMULA

LCM & HCF of Fractions

$$\text{LCM}\left(\tfrac{p}{q}, \tfrac{r}{s}\right) = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$$

Similarly, $\text{HCF of fractions} = \dfrac{\text{HCF of numerators}}{\text{LCM of denominators}}$.

Example: Fractions 3⁄8 and 5⁄12 → LCM(num)=LCM(3,5)=15; HCF(den)=HCF(8,12)=4 → LCM=15⁄4.

FORMULA

Euclidean Algorithm for HCF

$$\text{HCF}(a,b) = \text{HCF}(b,\ a \bmod b)$$

Replace the bigger number by the remainder until it becomes 0. The last non-zero remainder is the HCF.

Example: HCF(119, 34): 119 mod 34 = 17 → HCF(34,17)=17.

TRICK

LCM via Prime Factor Table

Write prime factorizations, take the *highest* power of each prime, multiply.

Useful when more than two numbers are involved.

EXAMPLE

Full Walk-through for 8, 12, 20

Prime factors: $8=2^3$, $12=2^2\times3$, $20=2^2\times5$.

LCM → highest powers: $2^3\times3\times5=120$.

HCF → lowest powers: $2^2=4$.

Prime Numbers

CONCEPT

What is Prime?

A natural number >1 whose only factors are 1 and itself. 1 is *not* prime by definition.

CONCEPT

First 25 Primes (Must Know List)

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

TRICK

√n Divisibility Test

To check if n is prime, test divisibility only by primes ≤ √n. If none divide, n is prime.

Example: 113; √113≈10.6 → test primes 2,3,5,7. None divide → 113 is prime.

TRICK **6k ± 1 Filter**

Every prime > 3 can be written as 6k − 1 or 6k + 1 (k integer). Speeds up option elimination.

CONCEPT **Co-prime (Relatively Prime)**

Two numbers whose HCF is 1. Example: 8 & 15. They need *not* both be prime.

CONCEPT **Twin Primes**

Pair of primes differing by 2 (e.g. 41 & 43). A favourite distractor in verbal questions on number properties.

Sum of Series

FORMULA **1 + 2 + … + n**

$$\text{Sum} = \frac{n(n+1)}{2}$$

Adds the first n natural numbers.

Example: 1 to 50 → 50×51/2 = 1275.

FORMULA **1² + 2² + … + n²**

$$\text{Sum} = \frac{n(n+1)(2n+1)}{6}$$

Used in variance, standard deviation or stair-step counting.

Example: n=5 → 5×6×11/6 = 55.

FORMULA **1³ + 2³ + … + n³**

$$\text{Sum} = \left[\frac{n(n+1)}{2}\right]^2$$

Equal to the square of the 1 + 2 + … + n formula. Handy mnemonic.

Example: n=4 → (4×5/2)² = 10² = 100.

FORMULA **Sum of an AP**

$$S_n = \frac{n}{2}\left(2a + (n-1)d\right)$$

a = first term, d = common difference, n = number of terms. If last term l known, use S_n=n(a+l)/2.

Example: For 5 terms of 7, 10, 13… → a=7, d=3, n=5 → S=5/2×(14+12)=5/2×26=65.

FORMULA

Sum of a Finite GP (r ≠ 1)

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

a = first term, r = common ratio, n = number of terms.

Example: 3 + 6 + 12 + 24 (n=4, r=2) → S=3(1−2⁴)/(1−2)=3(1−16)/(−1)=45.

FORMULA

Sum of First n Odd Numbers

$$1 + 3 + 5 + \ldots + (2n - 1) = n^2$$

Great shortcut for square numbers and pattern spotting.

Example: Sum of first 11 odd numbers = 11² = 121.

EXAMPLE

Identify Series Type Quickly

Series: 5, 15, 45, 135… Ratio constant (×3) → GP. So use GP sum, not AP. Spotting this difference is a common exam pitfall.

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CMAT · Formula Reference

Modern Math & DI

Permutations & Combinations

FORMULA **Factorial n!**

$$n! = n \times (n-1) \times (n-2) \times \ldots \times 1$$

What it means: n! (read “n factorial”) multiplies every positive integer from n down to 1.

Use it when: Counting ordered arrangements, permutations, or as a building block inside nPr and nCr.

Example: 5! = 5 × 4 × 3 × 2 × 1 = 120.

FORMULA **Permutations (order matters)**

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

Here n is the total number of distinct items, r is the number you arrange in *order*.

Apply when sequence matters (e.g. selecting 3 prize-winners for 1st, 2nd, 3rd).

Example: Ways to seat 4 out of 6 people in a row = 6P4 = 6!/2! = 720.

FORMULA **Combinations (order doesn’t matter)**

$$^{n}C_{r} = \frac{n!}{r!(n-r)!}$$

n = total distinct items, r = number you choose with *no* regard to order.

Use for team selection, subsets, lottery tickets, etc.

Example: Ways to choose 3 directors from 10 candidates = 10C3 = 120.

FORMULA **Circular permutations**

$$\text{Arrangements around a circle} = (n-1)!$$

n is the number of distinct items placed around a *circle*; rotations are considered identical.

Use for round tables, necklaces without flips.

Example: Ways to seat 5 friends around a round table = (5-1)! = 24.

FORMULA

Permutations with identical items

$$\text{Distinct arrangements} = \frac{n!}{p!\,q!\,r!\,\ldots}$$

n = total letters/objects. p, q, r … are counts of each set of identical items.

Choose when some objects are indistinguishable (MISSISSIPPI, colouring problems).

Example: ‘BANANA’ has 6 letters with A×3, N×2. Arrangements = 6! / (3! 2!) = 60.

CONCEPT

Slots (blank spaces) approach

Draw blank slots for each position, fill them step by step:

- Count choices for the first slot, then the next, multiplying as you go.
- Subtract or divide when restrictions apply (e.g. identical items, ‘together’ constraint).

Visualising slots avoids formula overload, especially in mixed restriction questions.

TRICK

Treat ‘together’ items as one block

If two or more items must remain adjacent, tie them into a single super-item first, arrange, then multiply by the internal arrangements.

Mini-example: Arrange the word ‘BOOK’ so the two O’s stay together: treat “OO” as one, you now have B, O O, K → 3! = 6 ways for blocks × 1! internal (O’s identical) = 6.

EXAMPLE

Example – restricted seating

Question: 6 boys and 4 girls sit in a row. How many ways if no two girls sit together?

1. Seat the 6 boys first → 6! ways.
2. This creates 7 gaps (| B | B | … | B |) where girls can be placed.
3. Select 4 of these 7 gaps for the girls: 7C4 ways.
4. Arrange the 4 girls in those chosen gaps: 4! ways.

Total = 6! × 7C4 × 4! = 720 × 35 × 24 = 604 800.

Probability

FORMULA

Basic probability

$$P(E) = \frac{\text{Favourable outcomes}}{\text{Total equally likely outcomes}}$$

P(E) reads “probability of event E”. The numerator counts ways the event happens; the denominator counts all possible outcomes, each assumed equally likely.

Example: Probability of drawing an ace from a standard deck = 4 / 52 = 1/13.

FORMULA

Complement rule

$$P(\text{not } E) = 1 - P(E)$$

"Not E" (also written E′) is everything outside event E.

Handy for "at least one", "none", or "not more than" style questions.

Example: If P(rain) = 0.3, then P(no rain) = 1 – 0.3 = 0.7.

FORMULA

Addition rule (mutually exclusive)

$$P(A \text{ or } B) = P(A) + P(B)$$

A and B cannot happen together (e.g. rolling a 2 or a 5 on one die).

Example: P(roll 2) = 1/6, P(roll 5) = 1/6 → P(2 or 5) = 1/6 + 1/6 = 1/3.

FORMULA

Addition rule (general)

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Subtracts overlap if A and B can occur together.

Example: In a class, 40% like coffee, 30% like tea, 10% like both. P(coffee or tea) = 0.4 + 0.3 – 0.1 = 0.6.

FORMULA

Multiplication rule (independent events)

$$P(A \text{ and } B) = P(A) \times P(B)$$

Use only when events don't influence each other (e.g. two separate coin tosses).

Example: P(head on 1st toss) × P(head on 2nd toss) = 0.5 × 0.5 = 0.25.

FORMULA

Conditional probability

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

Probability of A given B has already occurred.

Example: From a deck, given that the card drawn is red (P(B)=26/52=0.5), probability it's a heart (A and B) is 13/52 = 0.25. So P(heart | red) = 0.25 / 0.5 = 0.5.

TRICK

"At least one" shortcut

Instead of summing many cases, use complement:

P(at least one success) = 1 – P(no success at all).

Quick example: Toss a die 3 times. P(at least one 6) = $1 - (5/6)^3$ = 91/216.

EXAMPLE **Example – sum of two dice equals 8**

Total outcomes = 6 × 6 = 36 (ordered pairs).

Favourable pairs adding to 8: (2,6) (3,5) (4,4) (5,3) (6,2) = 5 cases.

P(sum = 8) = 5/36.

Set Theory

FORMULA **Union of two sets**

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

n(X) counts elements in X. Subtract the overlap to avoid double counting.

Example: 30 students play cricket, 25 football, 10 both → total playing at least one = 30 + 25 – 10 = 45.

FORMULA **Union of three sets**

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(AB) - n(BC) - n(CA) + n(ABC)$$

AB means A ∩ B, etc. Add singles, subtract pairwise overlaps, add back the triple overlap.

Example: If 40 like maths, 50 physics, 35 chemistry; 15 like both M&P, 10 P&C, 8 C&M, 5 like all three → total liking at least one = 40+50+35 – 15 – 10 – 8 + 5 = 97.

FORMULA **Complement of a set**

$$n(A^c) = n(U) - n(A)$$

U is the universal set (everything under discussion). A^c (not A) has whatever is left.

Example: In a survey of 120 people, 72 own cars. People without cars = 120 – 72 = 48.

CONCEPT **Reading Venn diagrams**

- Draw overlapping circles for each set.
- Fill the *deepest* overlap first (all three), then pair overlaps, then only-one regions, lastly people in none.
- Check that all region counts sum to the universal total.

TRICK **Exactly one of the sets**

For three sets:

Exactly-one = [Only A] + [Only B] + [Only C] = (singles) – 2×(pair overlaps) + 3×(triple) after working with the Venn regions. Filling the diagram numerically first is usually quicker than memorising a formula.

EXAMPLE

Example – language preferences

In a class of 60:

- Hindi speakers: 25
- English: 32
- Bengali: 18
- Hindi & English: 10
- English & Bengali: 7
- Hindi & Bengali: 5
- All three: 3

How many speak none of the three?

Total in at least one = 25+32+18 – 10 – 7 – 5 + 3 = 56.
None = 60 – 56 = 4.

Tabulation

FORMULA

Average from table values

$$\text{Average} = \frac{\text{Row or column total}}{\text{Number of observations}}$$

Pick the relevant row/column, sum its entries or read the provided total.

Example: Five quarterly profits: 8, 10, 9, 13, 10 → Average = 50 / 5 = ₹10 crore.

FORMULA

Percentage change

$$\%\,\text{Change} = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100\%$$

Applies to any two comparable table cells (sales this year vs last year, etc.).

Example: Sales rise from 1200 units to 1500 units → (1500 – 1200)/1200 × 100% = 25% rise.

FORMULA

Ratio of two quantities

$$\text{Ratio} = \frac{\text{Quantity A}}{\text{Quantity B}}$$

Always reduce to simplest form if the exam asks.

Example: Exports 240 mn, Imports 320 mn → Ratio Exports : Imports = 240/320 = 3 : 4.

CONCEPT

Reading a data table quickly

- Scan units (₹ lakh vs ₹ crore, tonnes vs kg).
- Circle/notate row and column totals if given.
- Underline what the question *really* asks (difference? %? highest?).

TRICK **The 10% mental anchor**

To estimate percentages without calc:

1. Find 10% quickly (move decimal one place).
2. Scale up or down: 5% = half of 10%, 25% = 2.5×10%, etc.

Speeds up all % change and contribution questions in tables.

EXAMPLE **Example – fastest growing region**

A table lists sales (₹ crore) for North region: 80 → 92 → 97 over three years. Percentage growth year-on-year:

- Year1→2: (92-80)/80 ×100% = 15%
- Year2→3: (97-92)/92 ×100% ≈ 5.43%

Largest growth is 15% in the first interval.

Pie Charts

FORMULA **Slice angle to actual value**

$$\text{Value of slice} = \frac{\text{Angle of slice}}{360^\circ} \times \text{Grand total}$$

Angle is measured in degrees. Grand total is the sum represented by the whole pie.

Example: If total expenditure is ₹600 mn and Marketing slice is 54°, value = 54/360 × 600 = ₹90 mn.

FORMULA **Value to percentage share**

$$\%\,\text{Share} = \frac{\text{Category value}}{\text{Grand total}} \times 100\%$$

Often quicker than converting via angles.

Example: Category sales ₹75 mn out of ₹300 mn → 75/300 ×100% = 25%.

FORMULA **Percentage to angle**

$$\text{Angle} = \%\,\text{Share} \times 3.6^\circ$$

Because 1% of 360° is 3.6°.

Example: 15% share → 15 × 3.6° = 54°.

CONCEPT **1% equals 3.6° rule**

Memorise common conversions:

- 10% → 36°
- 25% → 90° (a right angle)
- 50% → 180° (a straight line)
- 75% → 270°

TRICK **Comparing two slices quickly**

When exact numbers aren't required, compare angles visually or by subtracting their given degree measures—no need to find the total.

EXAMPLE **Example – infer grand total**

Pie shows R&D slice = 72°, labelled ₹180 mn.
Total = 180 mn ÷ (72/360) = 180 mn × 5 = ₹900 mn.

Bar Charts

FORMULA **Absolute difference between bars**

$$\text{Difference} = |\text{Value A} - \text{Value B}|$$

Use for questions like "How many more units did 2022 sell than 2021?"

Example: 2022 sales 680, 2021 sales 540 → Difference = |680 – 540| = 140 units.

FORMULA **Percentage difference (bar graph)**

$$\%\,\text{Increase} = \frac{\text{Later} - \text{Earlier}}{\text{Earlier}} \times 100\%$$

Same as in tables but often asked visually.

Example: Profit climbs from ₹40 lakh to ₹55 lakh → (55-40)/40 ×100% = 37.5% increase.

FORMULA **Stacked bar – part of total**

$$\%\,\text{Segment} = \frac{\text{Segment height}}{\text{Full bar height}} \times 100\%$$

Stacked bars show categories piled. Read each segment's value (often provided or can be deduced).

Example: Total bar 120 units tall, blue segment 30 → 30/120 ×100% = 25%.

CONCEPT **Always read the axis scale**

Check if the y-axis jumps (0, 20, 40, …) or uses broken scales. Misreading the scale is the #1 bar-chart trap.

TRICK **Eye-balling close bars**

When two bars look close, find difference by counting gridlines rather than reading exact numbers—saves time if only relative ranking is needed.

EXAMPLE **Example – highest CAGR from bar chart**

If bars show Company A revenue: 50 → 70 → 84 over 2 intervals: CAGR ≈ [(84/50)^(1/2) – 1] ×100% ≈ $(1.68^{0.5} - 1) \times 100\% \approx (1.296 - 1) \times 100\% \approx 29.6\%$.

Compare similarly for others to pick the highest.

Line Graphs

FORMULA **Slope (rate of change) between two points**

$$\text{Slope} = \frac{\text{Value}_2 - \text{Value}_1}{\text{Time}_2 - \text{Time}_1}$$

Measures how fast the variable changes over the interval.

Example: Production rises from 120 tons in 2019 to 180 tons in 2022 → Slope = (180-120)/(2022-2019) = 60/3 = 20 tons per year.

FORMULA **Compound Annual Growth Rate (CAGR)**

$$\text{CAGR} = \left(\frac{\text{Final}}{\text{Initial}}\right)^{\frac{1}{n}} - 1$$

n = number of years (or periods) between the two values.

Example: Value grows from 200 to 260 in 3 years → CAGR = (260/200)^(1/3) – 1 ≈ 1.099 – 1 = 0.099 = 9.9%.

FORMULA **Percentage change (any two points)**

$$\%\,\text{Change} = \frac{\text{Later} - \text{Earlier}}{\text{Earlier}} \times 100\%$$

Exactly the same computation you'd do in tables or bars.

Example: Visitors up from 4000 to 5200 → (5200-4000)/4000 ×100% = 30%.

CONCEPT **Handling multiple lines**

- Mark legend colours mentally.
- At any year, read vertically to find each line's value.
- Cross-overs answer "when did X overtake Y?" questions instantly.

TRICK **Mid-point approximation**

If the line looks straight between two labelled points, assume linearity and use the mid-point value = (V1 + V2)/2. Saves time for 'average during 2019-2021' type questions.

EXAMPLE **Example – forecast next year**

Line shows consistent +15 units per year (slope). Last shown year 2024 value = 180. Forecast 2025 → 180 + 15 = 195.

Data Sufficiency

CONCEPT **Standard answer choices**

1. Statement 1 alone is sufficient.
2. Statement 2 alone is sufficient.
3. Both together are sufficient, neither alone.
4. Each alone is sufficient.
5. Even both together are not sufficient.

CMAT/CAT style sometimes reorder them; read the directions first!

CONCEPT **Three-step evaluation**

1. **Focus on the question** – identify the exact value/relation required.
2. **Test Statement 1 alone** – assume only that data, ignore 2.
3. **Test Statement 2 alone**, then combine if necessary.

Stop testing as soon as you know the sufficiency category.

TRICK **Pick simple numbers to test**

When statements describe relations (e.g. x + y = 10), plug small integers to see if multiple results satisfy the main question. If two different plug-ins give different answers, the statement is *not* sufficient.

TRICK **No hidden mixing of statements**

While testing a statement alone, never import facts or variable definitions from the other statement. The examiners trap many students here.

EXAMPLE **Example – DS on triangle sides**

Question: What is the area of triangle ABC?

Stmt 1: Two sides are 5 cm and 12 cm, included angle 90°.
Stmt 2: Perimeter is 30 cm.

Analysis:
1 alone → right-angled triangle with legs 5 & 12, area = 0.5×5×12 = 30 cm² ⇒ Sufficient.
2 alone → Many triangles can have perimeter 30 ⇒ Not sufficient.
So answer is option 1.

CMAT · Formula Reference

Geometry & Mensuration

Lines

FORMULA **Distance between two points**

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

x_1, y_1 and **x_2, y_2** are coordinates of the two points.

Use it to measure straight-line (Euclidean) distance on a graph or map.

Example: From (2, 3) to (7, 11): √[(7-2)² + (11-3)²] = √[25 + 64] = √89 ≈ 9.43.

FORMULA **Internal section (ratio) formula**

$$\text{Point } P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

The line joining A(x_1, y_1) and B(x_2, y_2) is cut internally by point P in the ratio m : n (AP:PB).

Plug the numbers to locate the dividing point.

Example: Find 2:3 point between (0,0) and (10,0). x = (2×10 + 3×0)/(2+3) = 20/5 = 4, y = 0. So P(4, 0).

FORMULA **Slope of a line**

$$\text{Slope } (m) = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope m tells rise over run — steepness and direction.

Positive m = upward, negative m = downward, zero = horizontal, undefined = vertical.

Example: Line through (1, 2) and (5, 10): m = (10-2)/(5-1) = 8/4 = 2.

CONCEPT **Slope test for parallel/perpendicular**

- Two lines are **parallel** if their slopes are equal ($m_1 = m_2$).
- Two lines are **perpendicular** if $m_1 \times m_2 = -1$ (negative reciprocal).

FORMULA **Equation of a line through two points**

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

Start with any point (x_1, y_1). Plug slope from the two points to write the line in point-slope form.

Example: Through (3, 4) & (7, 6): slope = (6-4)/(7-3) = 2/4 = 1/2. Equation: y-4 = ½(x-3).

FORMULA **Mid-point of a segment**

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

The average of x-coordinates and y-coordinates gives the centre of the segment.

Example: Between (6, 1) and (2, 5): ((6+2)/2, (1+5)/2) = (4, 3).

Angles

FORMULA **Angles around a point**

$$\text{Sum} = 360^\circ$$

The total of all angles meeting at one point on a flat plane is 360°.

Example: If three rays create angles 110°, 95°, and x°, then x = 360 - (110+95) = 155°.

FORMULA **Linear pair**

$$\text{Angle A} + \text{Angle B} = 180^\circ$$

Two adjacent angles on a straight line add to 180°.

Example: One angle is 42°, the other is 180 - 42 = 138°.

CONCEPT **Vertically opposite angles**

When two straight lines cross, the opposite (X-shaped) angles are equal.

FORMULA **Angle sum of a triangle**

$$\text{Sum} = 180^\circ$$

Any triangle's three interior angles add to 180°.

Example: If two angles are 50° and 64°, the third is 66°.

FORMULA **Exterior angle of a triangle**

$$\text{Exterior} = \text{Remote interior 1} + \text{Remote interior 2}$$

The outer angle equals the sum of the two non-adjacent interior angles.

Example: Interior angles at A and B are 40° and 60°, exterior at C is 100°.

CONCEPT **Angles with parallel lines**

- **Corresponding** angles are equal.
- **Alternate interior** and **alternate exterior** are equal.
- **Co-interior (same-side interior)** sum to 180°.

FORMULA **Angle bisector ratio**

$$\frac{BD}{DC} = \frac{AB}{AC}$$

In triangle ABC, the bisector of ∠A meets BC at D.

The bisector divides the opposite side in the ratio of the adjacent sides.

Example: AB = 6 cm, AC = 9 cm ⇒ BD:DC = 6:9 = 2:3.

Triangles

FORMULA **Heron's area formula**

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

a, b, c are side lengths. s is the semi-perimeter: (a+b+c)/2.

Use when all three sides are known.

Example: Sides 7, 8, 9 ⇒ s = 12. Area = √[12×5×4×3] = √720 ≈ 26.83 sq units.

FORMULA **Area of right triangle**

$$\text{Area} = \frac{1}{2}\,(\text{Base})\,(\text{Height})$$

For a 90° triangle, the two perpendicular sides act as base and height.

Example: Legs 6 cm and 8 cm ⇒ Area = ½×6×8 = 24 cm².

FORMULA **Area of equilateral triangle**

$$\text{Area} = \frac{\sqrt{3}}{4}\,a^2$$

a is the common side length.

Example: a = 10 cm ⇒ Area ≈ 0.433×100 = 43.3 cm².

FORMULA **Pythagoras theorem**

$$\text{Hypotenuse}^2 = (\text{Perpendicular})^2 + (\text{Base})^2$$

Applies to right-angled triangles only.

Example: Perpendicular = 9, Base = 12 ⇒ Hypotenuse = √(81+144) = 15.

CONCEPT **Pythagorean triplets you must know**

Small	×2	×3
3 – 4 – 5	6 – 8 – 10	9 – 12 – 15
5 – 12 – 13	10 – 24 – 26	15 – 36 – 39
8 – 15 – 17	16 – 30 – 34	24 – 45 – 51

CONCEPT **Conditions for similarity**

- AA (two equal angles)
- SAS (two proportional sides with included equal angle)
- SSS (all sides proportional)

Corresponding sides are in equal ratio and corresponding angles are equal.

FORMULA **Area ratio of similar triangles**

$$\frac{\text{Area}_1}{\text{Area}_2} = \left(\frac{\text{Side}_1}{\text{Side}_2}\right)^2$$

For similar triangles, the ratio of areas is the square of the ratio of any pair of corresponding sides.

Example: Side ratio 2:3 ⇒ area ratio 4:9.

FORMULA **Area using in-radius**

$$\text{Area} = r \times s$$

r = in-radius (radius of inscribed circle). s = semi-perimeter.

Works for any triangle.

Example: r = 2 cm, s = 15 cm ⇒ Area = 30 cm².

FORMULA **Circum-radius of right triangle**

$$R = \frac{\text{Hypotenuse}}{2}$$

Only for right triangles; the hypotenuse is the diameter of the circumscribed circle.

Example: Hypotenuse = 10 cm ⇒ R = 5 cm.

FORMULA **Altitude from two sides and included angle**

$$\text{Height} = b \sin C$$

In triangle with sides a, b and included angle C between them, altitude on side a equals b sin C.

Example: b = 7 cm, C = 30° ⇒ Height = 7 × 0.5 = 3.5 cm.

Quadrilaterals

FORMULA **Sum of interior angles**

$$\text{Sum} = 360^\circ$$

All four interior angles of any quadrilateral add to 360°.

Example: Three angles 90°, 85°, 110° ⇒ fourth = 75°.

FORMULA

Area of a parallelogram

$$\text{Area} = \text{Base} \times \text{Height}$$

The height is the perpendicular distance between the two bases.

Example: Base = 12 cm, height = 5 cm ⇒ Area = 60 cm².

FORMULA

Area of a rectangle

$$\text{Area} = l \times b$$

l = length, b = breadth.

Example: 7 m × 4 m ⇒ 28 m².

FORMULA

Diagonal of a square

$$d = a\sqrt{2}$$

a is side length.

Example: a = 6 cm ⇒ d ≈ 8.49 cm.

FORMULA

Area of a rhombus (diagonals)

$$\text{Area} = \frac{1}{2}\, d_1\, d_2$$

d_1, d_2 are the diagonals.

Example: Diagonals 8 cm and 6 cm ⇒ Area = ½×8×6 = 24 cm².

FORMULA

Area of a trapezium

$$\text{Area} = \frac{1}{2}(a + b)h$$

a and b are the two parallel sides, h is the perpendicular height.

Example: a = 10 cm, b = 6 cm, h = 5 cm ⇒ Area = ½×16×5 = 40 cm².

CONCEPT

Opposite angles of cyclic quadrilateral

If all four vertices lie on a circle, each pair of opposite interior angles sum to 180°.

FORMULA

Area of a kite

$$\text{Area} = \frac{1}{2}\, d_1\, d_2$$

Same as rhombus: use product of diagonals.

Example: d_1 = 9 cm, d_2 = 4 cm ⇒ Area = 18 cm².

TRICK **Fastest perimeter check**

For rectangles with fixed area, perimeter is minimum when it is a square. Handy for max-min word problems.

Polygons

FORMULA **Sum of interior angles**

$$\text{Sum} = (n-2) \times 180^\circ$$

n = number of sides.

Example: For octagon (n = 8): (8-2)×180 = 1080°.

FORMULA **Each interior angle of regular polygon**

$$\text{Each} = \frac{(n-2) \times 180^\circ}{n}$$

Applicable when all sides and angles are equal.

Example: Regular hexagon: (6-2)×180/6 = 120°.

FORMULA **Each exterior angle (regular)**

$$\text{Each exterior} = \frac{360^\circ}{n}$$

Exterior angles of any convex polygon sum to 360°.

Example: Regular decagon (n = 10): each exterior = 360/10 = 36°.

FORMULA **Number of diagonals**

$$\text{Diagonals} = \frac{n(n-3)}{2}$$

n ≥ 3.

Example: n = 7 ⇒ 7×4/2 = 14 diagonals.

FORMULA **Area of regular polygon (using apothem)**

$$\text{Area} = \frac{1}{2} \text{Perimeter} \times \text{Apothem}$$

Apothem is the perpendicular distance from centre to a side.

Example: Regular pentagon: perimeter = 30 cm, apothem = 4 cm ⇒ Area = ½×30×4 = 60 cm².

TRICK **Exterior angle shortcut**

If you can spot one exterior angle of a regular polygon, sides n = 360° / exterior. Saves time over interior-angle equations.

Circles

FORMULA **Circumference**

$$C = 2\pi r$$

r = radius, π ≈ 3.1416.

Example: r = 7 cm ⇒ C ≈ 2×3.1416×7 ≈ 43.98 cm.

FORMULA **Area of circle**

$$A = \pi r^2$$

Same symbols as above.

Example: r = 10 m ⇒ A ≈ 314.16 m².

FORMULA **Area of a sector**

$$\text{Sector area} = \frac{\theta}{360^\circ}\,\pi r^2$$

θ = central angle in degrees.

Example: Quarter circle (θ = 90°), r = 8 cm ⇒ Area = 90/360×π×64 ≈ 50.27 cm².

FORMULA **Length of an arc**

$$\text{Arc} = \frac{\theta}{360^\circ}\,2\pi r$$

Same θ as sector.

Example: 60° arc, r = 5 cm ⇒ Arc = 60/360×2π×5 ≈ 5.24 cm.

FORMULA **Area of a segment (minor)**

$$\text{Segment} = \text{Sector area} - \frac{1}{2}r^2 \sin\theta$$

θ in radians or degrees but sine uses the same unit; many exams give degrees.

Example: θ = 60°, convert sin60° = 0.866. r = 10 cm. Sector = 60/360×π×100 ≈ 52.36. Triangle = ½×100×0.866 = 43.3. Segment ≈ 9.06 cm².

FORMULA **Chord length**

$$l = 2r \sin\frac{\theta}{2}$$

θ is the subtended central angle.

Example: θ = 120°, r = 6 cm ⇒ l = 2×6×sin60° = 12×0.866 ≈ 10.39 cm.

CONCEPT **Tangent-radius property**

A tangent meets the radius at 90°. Useful for right triangles involving tangents.

FORMULA **Power of a point (chord product)**

$$PA \times PB = PC \times PD$$

For two chords AB and CD intersecting at P, the product of the two parts of one chord equals that of the other.

Example: AP = 3 cm, PB = 6 cm, PC = 2 cm ⇒ PD = (3×6)/2 = 9 cm.

CONCEPT **Angle in same segment**

Angles made by the same chord at the circumference are equal.

TRICK **π substitution speed**

Unless options are symbolic, quickly pick π = 22/7 when radius is multiple of 7; else 3.14 is safer for decimals.

Mensuration (Area, Volume, Surface Area of 2D and 3D Figures)

FORMULA **Volume of cuboid**

$$V = l\,b\,h$$

l = length, b = breadth, h = height.

Example: 5 m × 4 m × 3 m ⇒ V = 60 m³.

FORMULA **Surface area of cube**

$$\text{Total SA} = 6a^2$$

a = edge length.

Example: a = 2 cm ⇒ SA = 6×4 = 24 cm².

FORMULA **Volume of cube**

$$V = a^3$$

Simple cube.

Example: a = 4 cm ⇒ V = 64 cm³.

FORMULA **Volume of cylinder**

$$V = \pi r^2 h$$

r = radius of base, h = height.

Example: r = 3 cm, h = 10 cm ⇒ V ≈ 282.74 cm³.

FORMULA **Curved and total surface area of cylinder**

$$\text{CSA} = 2\pi rh, \quad \text{TSA} = 2\pi rh + 2\pi r^2$$

CSA excludes top & bottom; TSA includes them.

Example: r = 5 cm, h = 7 cm ⇒ CSA ≈ 2π×5×7 ≈ 219.91 cm²; TSA ≈ 219.91 + 2π×25 ≈ 376.99 cm².

FORMULA **Volume of cone**

$$V = \frac{1}{3}\pi r^2 h$$

r = base radius, h = height.

Example: r = 4 cm, h = 9 cm ⇒ V ≈ 150.8 cm³.

FORMULA **Slant height of cone**

$$l = \sqrt{r^2 + h^2}$$

Pythagoras on radius and height.

Example: r = 3 cm, h = 4 cm ⇒ l = 5 cm.

FORMULA **Curved surface area of cone**

$$\text{CSA} = \pi rl$$

l = slant height.

Example: r = 3 cm, l = 5 cm ⇒ CSA ≈ 47.12 cm².

FORMULA **Volume of sphere**

$$V = \frac{4}{3}\pi r^3$$

Perfect ball.

Example: r = 6 cm ⇒ V ≈ 904.78 cm³.

FORMULA **Surface area of sphere**

$$\text{SA} = 4\pi r^2$$

No edges.

Example: r = 5 cm ⇒ SA ≈ 314.16 cm².

FORMULA **Hemisphere areas & volume**

$$\text{CSA} = 2\pi r^2,\ \text{TSA} = 3\pi r^2,\ V = \frac{2}{3}\pi r^3$$

CSA excludes base circle; TSA includes it.

Example: r = 7 cm ⇒ CSA ≈ 308 cm², TSA ≈ 461.58 cm², V ≈ 718.4 cm³.

FORMULA

Volume of frustum of cone

$$V = \frac{1}{3}\pi h(r_1^2 + r_2^2 + r_1 r_2)$$

r_1, r_2 are radii of the two circular faces, h is the vertical height.

Example: r_1 = 4 cm, r_2 = 2 cm, h = 6 cm ⇒ V ≈ 1/3×π×6×(16+4+8) = 2π×28 ≈ 175.93 cm³.

TRICK

Unit conversion ladder

- 1 m = 100 cm, 1 cm = 10 mm
- Area: multiply square of factor (1 m² = 10 000 cm²)
- Volume: multiply cube of factor (1 m³ = 1 000 000 cm³)

EXAMPLE

Quick paint requirement

Problem: How much paint (1 L covers 5 m²) is needed to coat the outside of a cylindrical tank of radius 2 m and height 3 m (top not painted)?

Surface to paint = CSA = 2πrh = 2×π×2×3 ≈ 37.70 m².

Paint required = 37.70 / 5 ≈ 7.54 L ⟹ buy 8 L.

cracku

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Clocks & Calendars

Time-angle relationships

CONCEPT **What moves how much?**

A clock dial is a full circle of 360°. It is marked into 12 equal hour divisions, so each hour mark is 30° apart (360° ÷ 12). The minute marks (60 of them) are 6° apart (360° ÷ 60).

- Minute hand speed: 6° every minute.
- Hour hand speed: 0.5° every minute (30° an hour).
- Second hand (rarely asked): 6° every second.

FORMULA **Angle moved by minute hand**

$$\text{Angle} = 6 \times M$$

M is the number of minutes counted from 12 o'clock (0 min). Works for any fraction too: for 15.5 minutes use 15.5.

Example: 20 minutes past any hour ⇒ Angle = 6 × 20 = 120° from the 12 o'clock position.

FORMULA **Angle moved by hour hand**

$$\text{Angle} = 30 \times H + 0.5 \times M$$

H is the hours completed after 12 (0–11), M is the additional minutes.
Use when the hour hand has crept between hour marks.

Example: At 4:30, H = 4, M = 30 ⇒ Angle = 30 × 4 + 0.5 × 30 = 120 + 15 = 135° from 12.

FORMULA **Angle between hour and minute hands**

$$\text{Angle} = |30 \times H - 5.5 \times M|$$

H is the hour number (1–12), M is minutes past that hour. The absolute value | | makes the result positive. If the answer exceeds 180°, subtract from 360° for the smaller angle usually asked.

Example: At 5:40, H = 5, M = 40 ⇒ |30×5 − 5.5×40| = |150 − 220| = 70°. Hands are 70° apart.

CONCEPT **Relative speed of the two hands**

Minute hand gains on hour hand at 5.5° every minute (6° − 0.5°). This single figure drives most "meet" or "overtake" problems.

FORMULA **How many times do the hands coincide?**

> Coincidences in 12 h = 11

Because the minute hand must lap the hour hand once every 65 4⁄11 minutes (360° ÷ 5.5°). Therefore 11 laps in 12 hours and 22 in a full day.

Quick check: Between 12 noon and 12 midnight, the last meeting is at ≈ 10:55 pm; they do not meet exactly at 12 again within the same 12-hour stretch.

TRICK **Mirror or opposite times**

For a time T on a normal clock, the mirror time T′ satisfies *hour + minute/60 = 11* (for 12-hour mirrors). Fast shortcut: subtract the given time from 11:60.

Example: Mirror image of 3:37 → 11:60 − 3:37 = 8:23.

EXAMPLE **When are the hands together after 4 o'clock?**

Use relative speed 5.5°/min. At 4 pm the hour hand is at 120° (4×30). Let t minutes pass for coincidence.

Angle gained by minute hand = 5.5 t. Need 360° multiple difference ⇒ 5.5 t = 120° (to catch up one full lap ahead isn't needed because minute starts at 0°).
t = 120 ÷ 5.5 ≈ 21 49⁄11 min ≈ 21 min 49 s.

Answer: 4:21 : 49 (approximately).

Odd days calculations

CONCEPT **What is an odd day?**

Count total days elapsed, divide by 7, and keep only the remainder. That remainder (0–6) is the number of **odd days**. It tells how far we have moved forward in the weekday cycle.

Example: 10 days → 10 ÷ 7 = 1 week + 3 odd days; so a Monday plus 10 days lands on Thursday.

FORMULA **Odd days in full centuries (Gregorian)**

Century	1600/2000	1700/2100	1800/2200	1900/2300
Odd days	0	5	3	1

Take the first two digits of the year, match to the table, and use the listed odd-day count.

Example: Year 1984 → century part 19 ⇒ 1 odd day contributed by 1900–1999 block.

FORMULA **Odd days in each month**

Mon	*Jan*	*Feb*	*Mar*	*Apr*	*May*	*Jun*	*Jul*	*Aug*	*Sep*	*Oct*	*Nov*	*Dec*
Normal	3	0	3	2	3	2	3	3	2	3	2	3
Leap	3	1	3	2	3	2	3	3	2	3	2	3

Values are the odd days each month adds. Use the 'Leap' row only if the year is a leap year (divisible by 4 but not by 100, unless also by 400).

Tip: Memorise Jan=3, May=3, Jul=3, others mostly 2 or 0/1.

TRICK **Is it a leap year?**

- Divisible by 400 → Leap (e.g., 2000).
- Otherwise divisible by 100 → Not leap (e.g., 1900).
- Otherwise divisible by 4 → Leap (e.g., 2024).
- Else → Not leap.

FORMULA **Day of the week from a date**

$$\text{Weekday index} = (C + Y + L + M + D) \bmod 7$$

C = century odd days, Y = odd days from years within the century, L = leap day additions (number of leap years completed within Y), M = month code (from table), D = date number. Take the sum, divide by 7, remainder 0–6 maps to Sunday–Saturday.

Example below shows the steps.

EXAMPLE **15 August 1947 – which day?**

1. **Century (19xx):** C = 1 odd day.
2. **Years inside century:** 47 years add 47 odd days.
   Leap years among them = $\lfloor 47 \div 4 \rfloor = 11$ → L = 11 odd days.
3. Reduce Y + L: (47 + 11) = 58 → 58 ÷ 7 = 8 weeks + 2 odd days.
4. **Month:** August normal year → M = 3.
5. **Date:** D = 15.
6. Total odd days = C 1 + (Y+L) 2 + M 3 + D 15 = 21.
7. 21 ÷ 7 = 0 remainder ⇒ Weekday index 0 ⇒ **Friday**.

(Indeed, India's Independence Day was a Friday.)

TRICK **Quick anchor: 1 Jan 2000 was Saturday**

Many recent-year questions span 20xx dates. Remember that 1 Jan 2000 fell on a Saturday (odd day 6). Using it instead of the century table often speeds up mental work.

Calendar repeating years

CONCEPT **When does a calendar repeat?**

A later year shares an identical calendar if (a) both years are of the same type (leap or non-leap) and (b) the gap between them shifts weekdays by a multiple of 7.

FORMULA **Weekday shift caused by a year**

$$\text{Shift} = \begin{cases} 1 & \text{for a normal year} \\ 2 & \text{for a leap year} \end{cases}$$

A normal year (365 days) leaves 1 odd day; a leap year (366 days) leaves 2. Add shifts cumulatively until the total is a multiple of 7 to find the next repeat.

Example: From 2019 (normal) → 2020 adds 1, 2020 (leap) adds 2, 2021 adds 1… keep counting.

TRICK

Shortcut for normal years

For most non-leap years, the calendar repeats after 6 years (1+1+2+1+1) = 6 odd days → still not multiple of 7, so add one more normal year: 6 + 1 = 7 odd days.

Exceptions: When a leap year intervenes earlier (e.g., years just before a leap year).

CONCEPT

28-year master cycle

In the Gregorian calendar, the exact day-date pattern returns every 28 years (7 days × 4 year-types pattern). Handy estimate when searching far ahead or behind.

EXAMPLE

Find the next year with the same calendar as 2019

2019 is a normal year.

1. 2020 (leap) shift +2 → cumulative 2.
2. 2021 shift +1 → 3.
3. 2022 shift +1 → 4.
4. 2023 shift +1 → 5.
5. 2024 (leap) shift +2 → 7 (multiple of 7).

Answer: 2025 shares the same calendar as 2019.

TRICK

Leap year calendars repeat after 28 or sometimes 12 years

A leap year repeats after 28 years, *unless* the next leap year starts on the same weekday (gap 12 years: 28 minus 16). Quick litmus: If the leap year starts on a Monday or Wednesday, it repeats in 28; if it starts on a Saturday, it can repeat in 12.

FORMULA

Number of leap years between two years

$$\text{Leaps} = \left(\left\lfloor \frac{Y_2}{4} \right\rfloor - \left\lfloor \frac{Y_2}{100} \right\rfloor + \left\lfloor \frac{Y_2}{400} \right\rfloor \right) - \left(\left\lfloor \frac{Y_1 - 1}{4} \right\rfloor - \left\lfloor \frac{Y_1 - 1}{100} \right\rfloor + \left\lfloor \frac{Y_1 - 1}{400} \right\rfloor \right)$$

This counts leap years from year 1 to Y_2 and subtracts count up to $Y_1 - 1$. Useful when checking larger spans for repeating calendars.

Example: Between 2001 and 2025 inclusive there are 6 leap years (2004, 08, 12, 16, 20, 24).

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Arithmetic

Percentages

CONCEPT **What a percent really is**

Percent literally means “per 100”. Writing 37% is a shorthand for saying 37 parts out of 100 parts of the same whole. All percentage questions are nothing but fraction questions with denominator = 100.

FORMULA **Percent to fraction (and back)**

$$\text{Percent as fraction} = \frac{p}{100} \qquad \text{and} \qquad \text{Fraction } \frac{a}{b} = \left(\frac{a}{b} \times 100\right)\%$$

Here p is the percentage value, a/b any ordinary fraction.

Use whenever you need to switch forms quickly.

Example: 12.5% = 12.5 / 100 = 1 / 8. Conversely, 3 / 5 = 0.6 × 100 = 60%.

FORMULA **Percentage change (increase or decrease)**

$$\%\,\text{Change} = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100\%$$

“Old” is the starting value, “New” is the value after change.

Apply whenever you compare a before–after pair.

Example: Price rises from ₹240 to ₹300. % Change = (300 − 240)/240 × 100 = 25% increase.

FORMULA **Direct multiplier for one change**

$$\text{Final} = \text{Original} \times \left(1 \pm \frac{k}{100}\right)$$

k is the percentage increase (+) or decrease (−).

Speeds up “after 18% discount” or “after 12% GST” calculations.

Example: ₹4,000 after 15% discount: 4,000 × (1 − 0.15) = ₹3,400.

TRICK **Two successive percentages in 3-seconds**

If a value changes by a% and then by b%, the net % change is:
a + b + (ab/100) (take signs: + for rise, – for fall).

Example: Mark-up 20% then discount 10% ⇒ Net = 20 + (-10) + (20×-10)/100 = 8% rise overall.

FORMULA **Finding original from final (reverse %)**

$$\text{Original} = \frac{\text{Final}}{1 \pm k/100}$$

Use when the percentage change and final value are known but the original is asked.

Example: Salary after 25% hike is ₹50,000. Original = 50,000 / 1.25 = ₹40,000.

CONCEPT **Memory table: killer common conversions**

Fraction	Percent
1/2	50%
1/3	33⅓%
1/4	25%
1/5	20%
3/4	75%
5/6	83⅓%

Know these by heart to slash calculation time.

EXAMPLE **Full worked problem – successive discount**

A shirt is marked ₹1,200. A store gives 10% festival discount followed by an additional 5% card discount. What is the selling price?

1. Net % change = -10 + (-5) + (-10×-5)/100 = -14.5%.
2. Selling price = 1,200 × (1 - 0.145) = 1,200 × 0.855 = ₹1,026.

Profit and Loss

CONCEPT **Key terms: CP, SP, MP, Discount**

- **Cost Price (CP):** What the trader paid.
- **Selling Price (SP):** What the customer pays.
- **Marked/List Price (MP):** Label price before any discount.
- **Discount:** MP − SP.

FORMULA **Profit, Loss and their percentages**

$$\text{Profit} = \text{SP} - \text{CP},\ \text{Loss} = \text{CP} - \text{SP}\ \text{Profit}\% = \frac{\text{Profit}}{\text{CP}} \times 100\%,\ \text{Loss}\% = \frac{\text{Loss}}{\text{CP}} \times 100\%$$

Base is always CP unless stated otherwise.

Example: Buy at ₹800, sell at ₹880 ⇒ Profit 80, Profit% = 80/800 × 100 = 10%.

FORMULA **Shortcut multipliers**

$$\text{SP} = \text{CP} \times \left(1 + \frac{p}{100}\right), \qquad \text{CP} = \frac{\text{SP}}{1 + p/100}$$

p is profit% (take negative p for loss).

Apply when margin and cost (or selling) price are given.

Example: 15% gain on ₹400 cost ⇒ SP = 400 × 1.15 = ₹460.

FORMULA **Discount percentage**

$$\text{Discount}\% = \frac{\text{MP} - \text{SP}}{\text{MP}} \times 100\%$$

Base is Marked Price.

Example: MP ₹1,000, sold for ₹850 ⇒ Discount% = 150/1,000 × 100 = 15%.

TRICK **Two successive discounts**

Net discount% = a + b − (ab/100). Works exactly like successive % decrease.

Example: 20% + 10% discounts ⇒ 20 + 10 − 2 = 28% net discount.

FORMULA **Needed mark-up for desired profit after discount**

$$\text{Required MP} = \frac{\text{CP} \times \left(1 + \dfrac{\text{Desired Profit}\%}{100}\right)}{1 - \dfrac{\text{Discount}\%}{100}}$$

Helps reverse-engineer price tags.

Example: CP ₹600, shop wants 25% profit after giving 10% discount.
MP = 600×1.25 / 0.9 = ₹833.33 → round up to ₹834.

EXAMPLE **Chain transaction (dealer + retailer)**

A wholesaler sells goods to a retailer at 10% profit on CP. The retailer marks them 30% above his cost and allows 20% discount to customers. Find retailer's profit% on his cost.

Retailer's MP multiplier = 1.30. After 20% discount, SP to customer = 1.30 × 0.80 = 1.04 times retailer's CP. Thus profit% = 4%.

Simple and Compound Interest

FORMULA **Simple interest**

$$\text{SI} = \frac{P\,R\,T}{100}, \qquad \text{Amount} = P + \text{SI}$$

P = principal (initial money), R = annual rate %, T = time in years.

Example: ₹10,000 at 8% for 3 y ⇒ SI = 10,000×8×3/100 = ₹2,400, Amount = ₹12,400.

FORMULA

Compound interest (interest added yearly)

$$\text{Amount} = P\left(1 + \frac{R}{100}\right)^{T}$$

Same symbols as above. Compound Interest = Amount − P.

Example: ₹5,000 at 10% for 2 y ⇒ Amount = 5,000×1.1² = 5,000×1.21 = ₹6,050. CI = 1,050.

FORMULA

Compounded m times a year

$$\text{Amount} = P\left(1 + \frac{R}{100m}\right)^{mT}$$

m = number of compounding periods per year (2 for half-yearly, 4 for quarterly, etc.).

Example: ₹2,000 at 12% compounded quarterly for 1 y ⇒ m = 4. Amount = 2,000×(1+0.12/4)^4 = 2,000×(1.03)^4 ≈ ₹2,000×1.1255 ≈ ₹2,251.

FORMULA

Difference between CI and SI for 2 years

$$\text{CI} - \text{SI} = P\left(\frac{R}{100}\right)^{2}$$

Handy shortcut for exactly 2 years, yearly compounding.

Example: P = ₹6,000, R = 10% ⇒ Diff = 6,000×(0.1)² = ₹60.

TRICK

Effective annual rate when compounding more often

Use formula in previous block then subtract 1 and multiply by 100 to get effective %.

Example: 12% compounded half-yearly ⇒ (1+0.12/2)² − 1 = 1.06² − 1 = 1.1236 − 1 = 0.1236 ⇒ 12.36% effective.

CONCEPT

Rule of 72

Time to double = 72 / R (R in % per annum, compound interest). Approximate but good for mental checks.

EXAMPLE

Loan comparison example

Bank A offers 14% SI, Bank B offers 12% CI yearly. For 3 years on ₹50,000 which is cheaper?

- Bank A: SI = 50,000×14×3/100 = ₹21,000 ⇒ Pay 71,000.
- Bank B: CI Amount = 50,000×1.12³ ≈ 50,000×1.4049 = ₹70,245.

Bank B is cheaper by about ₹755.

Time, Speed & Distance

FORMULA **Speed, distance, time triangle**

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}},\ \text{Distance} = \text{Speed} \times \text{Time},\ \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

Units must be compatible (e.g. km with hours).

Example: 150 km in 3 h ⇒ Speed = 150/3 = 50 km/h.

CONCEPT **km/h ↔ m/s quick convert**

Multiply km/h by 5/18 to get m/s. Multiply m/s by 18/5 to get km/h.

Example: 72 km/h = 72×5/18 = 20 m/s.

FORMULA **Average speed for equal distances**

$$\text{Average Speed} = \frac{2ab}{a+b}$$

Two legs of equal distance covered at speeds a and b.

Example: 60 km/h going, 40 km/h return ⇒ Avg = 2×60×40 / (60+40) = 48 km/h.

FORMULA **Relative speed of two bodies**

$$\text{Relative Speed} = \begin{cases} a+b & \text{opposite directions} \\ |a-b| & \text{same direction} \end{cases}$$

a and b in same units. Use for trains, overtaking, stream problems.

Example: Trains at 50 km/h and 70 km/h opposite ⇒ Rel speed 120 km/h.

FORMULA **Time for train to cross pole or platform**

$$\text{Time}_{\text{pole}} = \frac{L}{S}, \qquad \text{Time}_{\text{platform}} = \frac{L+P}{S}$$

L = train length, P = platform length, S = train speed.

Example: 200 m train at 20 m/s crosses 100 m platform ⇒ Time = (200+100)/20 = 15 s.

FORMULA **Speed of boat in still water and stream**

$$\text{Downstream} = b+s, \qquad \text{Upstream} = b-s$$

b = boat's still-water speed, s = stream speed. Solve two equations to find b and s when journey times are given.

Example: Downstream 12 km in 1 h (12 km/h), Upstream same 12 km in 1.5 h (8 km/h) ⇒ b = (12+8)/2 = 10 km/h, s = 2 km/h.

TRICK **Meeting point ratio shortcut**

When two objects start simultaneously toward each other, the distances travelled are in the inverse ratio of their speeds.

Example: Speeds 30 km/h and 50 km/h ⇒ They meet closer to the slower one in ratio 50:30 = 5:3.

Time and Work

FORMULA **Individual work rate**

$$\text{Rate} = \frac{1}{a} \text{ work per day}$$

a = days a person/machine takes to finish one job alone.

Example: If A finishes job in 5 d ⇒ Rate = 1/5 job per day.

FORMULA **Combined work time for two workers**

$$\frac{1}{T} = \frac{1}{a} + \frac{1}{b}$$

a and b are individual times, T is time together.

Example: 6 d and 10 d ⇒ 1/T = 1/6+1/10 = 8/30 ⇒ T = 30/8 = 3.75 d.

TRICK **LCM method – make numbers friendly**

Assume total work = LCM of given days (or any convenient multiple). Convert each worker's rate to whole units, then proceed with simple arithmetic. Eliminates fractions.

FORMULA **Efficiency and time relation**

$$\text{Efficiency} \propto \frac{1}{\text{Time}}$$

Higher efficiency ⇒ fewer days. If A is twice as efficient as B, A's time = half of B's.

Example: A finishes in 9 d, B in 15 d. Efficiency ratio = 15:9 = 5:3.

CONCEPT **Pipes & cistern = positive and negative work**

Filling pipe: positive rate. Leakage/emptying pipe: negative rate. Sum rates exactly like worker problems.

EXAMPLE **Switching workers mid-way**

A can do a job in 8 d. After he worked 3 d, B joined and finished in 2 more days. How long would B alone take?

1. Total work = 1 job. A's rate = 1/8.
2. Work done by A in 3 d = 3/8.
3. Remaining work = 5/8 completed in 2 d together ⇒ Combined rate = (5/8)/2 = 5/16.
4. B's rate = 5/16 − 1/8 = 5/16 − 2/16 = 3/16.
5. B alone time = 1 ÷ (3/16) = 16/3 = 5⅓ d.

Ratio and Proportion

CONCEPT **Ratio basics**

a : b expresses the relative sizes of a and b, not absolute values. Multiply or divide both terms by the same positive number and the ratio stays the same.

FORMULA **Proportion property**

$$a:b = c:d \iff ad = bc$$

Cross-product test. Use to find missing term.

Example: 3:x = 9:12 ⇒ 3×12 = 9x ⇒ x = 4.

FORMULA **Fourth proportional**

$$\text{Fourth } d = \frac{b\,c}{a} \text{ when } a : b = c : d$$

Three numbers known, find the fourth.

Example: 4:6 = 10:d ⇒ d = 6×10/4 = 15.

CONCEPT **Compound ratio**

Multiply corresponding terms of given ratios. (a:b)×(c:d) ⇒ ac : bd.

Useful in two-step mixture problems.

FORMULA **Dividing a quantity in a ratio**

$$\text{Share of A} = \frac{a}{a+b} \times \text{Total}, \quad \text{Share of B} = \frac{b}{a+b} \times \text{Total}$$

a:b is the ratio, Total the quantity.

Example: Divide ₹900 in 2:1 ⇒ A: 2/3×900=₹600, B: ₹300.

CONCEPT **Direct vs inverse proportion**

Direct: increase in one ➔ proportional increase in other (e.g. cost ∝ quantity). Inverse: product stays constant (e.g. speed × time = distance fixed).

EXAMPLE **Mixture replacement example**

A container has milk and water in 7:3 ratio. If 10 L is removed and replaced by water, new ratio is 5:4. Find initial quantity.

Let initial volume = 10x. Milk = 7x, Water = 3x.
Milk left after removal = 7x − 7/10×10 = 7x − 7.
Total after refill = 10x.
Now milk : water = 5:4 ⇒ Milk = 5/9×10x = 50x/9.
So 7x − 7 = 50x/9 ⇒ 63x − 63 = 50x ⇒ 13x = 63 ⇒ x = 4.846. Initial volume ≈ 48.5 L.

Averages

FORMULA **Simple average**

$$\text{Average} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Applies to marks, ages, runs, anything additive.

Example: (15, 17, 20) ⇒ Avg = (15+17+20)/3 = 52/3 ≈ 17.33.

FORMULA **Weighted average**

$$\text{Weighted Avg} = \frac{\sum w_i x_i}{\sum w_i}$$

x_i are values, w_i their weights (credits, quantities).

Example: Mix sugar at ₹30/kg (2 kg) and ₹45/kg (3 kg): Avg price = (30×2+45×3)/(2+3)=(60+135)/5=₹39.

FORMULA **Average after adding/removing member**

$$\text{New Sum} = \text{Old Avg} \times n \ \pm \ \text{Value added/removed}$$

Then divide by new count to get new average.

Example: Class avg 70 for 20 students. One new student scores 90. New avg = (70×20+90)/21 = 71.0.

TRICK **Alligation cross method**

Place cheaper and dearer values on left/right, mean in middle. Cross-subtract to get mixing ratio in one line. Saves time in mixture problems.

CONCEPT **Running (cumulative) average**

Update average quickly: New avg = Old avg + (New value − Old avg)/n_{new}. Great for batting average questions.

CONCEPT **Average of ratios ≠ ratio of averages**

Don't simply average speeds, percentages, or ratios unless *weights* (distances, totals) are equal. Always check the base.

EXAMPLE **Cricket batting average drop**

A player's average after 20 innings is 42. After next innings, his average drops to 41. Find runs in 21st innings.

Old sum = 42×20 = 840. New average 41 over 21 innings ⇒ New sum = 41×21 = 861. Runs scored = 861 − 840 = 21.

CMAT · Formula Reference

Algebra

Linear and Quadratic Equations

FORMULA Solution of a single linear equation

$$x = \frac{-b}{a}$$

For a linear equation of the form **ax + b = 0**,
a is the coefficient of x, b is the constant term.

Use whenever only one unknown appears to the first power.

Example: 3x + 9 = 0 ⇒ x = −9/3 = −3.

FORMULA Two linear equations in two unknowns (elimination shortcut)

$$x = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \quad y = \frac{a_2c_1 - a_1c_2}{a_1b_2 - a_2b_1}$$

The pair is
(1) $a_1x + b_1y = c_1$
(2) $a_2x + b_2y = c_2$.

Apply when quick elimination by cross-multiplying is easier than substitution.

Example: 2x + 3y = 13, 5x − y = 9.
Denominator = 2(−1) − 5(3) = −2 − 15 = −17.
x = (3·9 − (−1)·13)/(−17) = (27 + 13)/(−17) = 40/−17 = −40⁄17.
y = (5·13 − 2·9)/(−17) = (65 − 18)/(−17) = 47/−17 = −47⁄17.

FORMULA Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solves $ax^2 + b\,x + c = 0$ for x.
Discriminant $D = b^2 − 4ac$ decides nature of roots.

Example: $x^2 − 5x + 6 = 0$ ⇒ D = 25 − 24 = 1.
x = (5 ± 1)/2 ⇒ x = 3 or 2.

FORMULA Sum and product of quadratic roots

$$\text{Sum of roots} = -\frac{b}{a},\ \text{Product of roots} = \frac{c}{a}$$

For $ax^2 + bx + c = 0$ with roots r_1, r_2.

Example: $2x^2 + 7x + 3 = 0$.
Sum = $-7/2 = -3.5$, Product = $3/2 = 1.5$.

FORMULA Quadratic equation from given roots

$$x^2 - (\text{sum})x + (\text{product}) = 0$$

After finding sum = $r_1 + r_2$ and product = $r_1 r_2$, plug in.

Example: Roots 4 and $-1 \rightarrow$ sum = 3, product = -4.
Equation: $x^2 - 3x - 4 = 0$.

CONCEPT Discriminant nature of roots

- $D > 0$: two distinct real roots.
- $D = 0$: one repeated real root.
- $D < 0$: pair of complex (non-real) roots – rarely tested in CMAT.

TRICK Quick quadratic spotting

- **Perfect square?** If $b^2 = 4ac$, the quadratic is a perfect square: $(\sqrt{a}\,x + \sqrt{c})^2$.
- **Middle-term factorisation:** Split b into two numbers whose sum is b and whose product is ac.
- **Free term test:** For integer roots, they must divide c (if a = 1) or c/a (after making the equation monic).

EXAMPLE Elimination in 30 seconds

Solve $3x + 2y = 17$ and $5x - 2y = 11$.

1. Add the equations (coefficients of y cancel): $8x = 28 \Rightarrow x = 3.5$.
2. Plug back into $3x + 2y = 17 \Rightarrow 3 \cdot 3.5 + 2y = 17 \Rightarrow 10.5 + 2y = 17 \Rightarrow 2y = 6.5 \Rightarrow y = 3.25$.

Tip: Opposite coefficients? Add. Same sign? Subtract.

Inequalities

FORMULA Solving a simple linear inequality

$$\text{If } a > 0,\ ax + b < c \Rightarrow x < \frac{c - b}{a}$$

Works exactly like solving $ax + b = c$, except the inequality sign stays the same because we divide by a positive number.

Example: $3x + 4 < 13 \Rightarrow x < (13 - 4)/3 = 3$.

CONCEPT

Sign-flip rule

When you multiply or divide *both sides* of an inequality by a negative number, the inequality symbol flips.

Example: −2x > 6 ⇒ divide by −2 ⇒ x < −3 (note the change from > to <).

FORMULA

Three-part (sandwich) inequality

$$a < bx + c \leq d \implies \frac{a-c}{b} < x \leq \frac{d-c}{b}$$

a, b, c, d are constants, b positive. Solve the two ends separately.

Example: 2 < 3x + 1 ≤ 7 ⇒ subtract 1: 1 < 3x ≤ 6 ⇒ divide 3: 1/3 < x ≤ 2.

FORMULA

Quadratic inequality via sign chart

$ax^2 + bx + c$ changes sign only at its roots

Factor $ax^2 + bx + c = a(x - r_1)(x - r_2)$. Mark the roots on a number line; the parabola is above the x-axis (positive) outside the roots if a > 0, and below it (negative) between the roots. Reverse for a < 0.

Example: Find x where x² − 5x + 6 > 0.
Roots 2 and 3. Since a = 1 > 0, expression is positive for x < 2 or x > 3.

FORMULA

Basic modulus inequality

$$|x - a| < k \implies a - k < x < a + k$$

|x − a| measures distance of x from a. The inequality says that distance is less than k.

Example: |x − 5| < 2 ⇒ 3 < x < 7.

FORMULA

AM ≥ GM for two numbers

$$\frac{p+q}{2} \geq \sqrt{pq}$$

p and q are non-negative numbers. Equality holds only when p = q.

Example: Numbers 8 and 18 → AM = 13, GM = √(144) = 12; indeed 13 ≥ 12.

CONCEPT

Open vs closed intervals

- (a, b): excludes a and b → inequality uses <, >.
- [a, b]: includes both ends → ≤, ≥.
- Mixed brackets like (a, b] possible.

TRICK

Sign chart in 3 steps

1. Factor expression completely.
2. Plot critical points (where any factor is zero) on a line.
3. Pick one test value in each interval; once you know the sign there, it is the sign for that whole interval.

Much faster than plugging many random numbers.

Algebraic Expressions

FORMULA **(a + b)²**

$$(a+b)^2 = a^2 + 2ab + b^2$$

Expands a binomial squared into three easy terms.

Example: (7 + 3)² = 49 + 2·7·3 + 9 = 49 + 42 + 9 = 100.

FORMULA **(a − b)²**

$$(a-b)^2 = a^2 - 2ab + b^2$$

Same pattern as (a + b)² with the middle term negative.

Example: (10 − 4)² = 100 − 80 + 16 = 36.

FORMULA **(a + b)(a − b)**

$$(a+b)(a-b) = a^2 - b^2$$

Transforms a difference of squares into a product and vice-versa – great for mental multiplication.

Example: 23² − 17² = (23 + 17)(23 − 17) = 40·6 = 240.

FORMULA **(a + b)³**

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Useful in expansion and factorisation questions.

Example: (2 + 3)³ = 8 + 36 + 54 + 27 = 125.

FORMULA **a³ + b³ factorisation**

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

Turns a sum of cubes into an easily factorable product.

Example: 8 + 27 = (2 + 3)(4 − 6 + 9) = 5·7 = 35.

FORMULA **a³ − b³ factorisation**

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Mirror image of the sum-of-cubes formula.

Example: 64 − 27 = (4 − 3)(16 + 12 + 9) = 1·37 = 37.

FORMULA **(a + b + c)²**

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

Remember: square each term, then double every distinct pair.

Example: (1 + 2 + 3)² = 1 + 4 + 9 + 4 + 12 + 6 = 36.

TRICK **Spotting hidden identities**

If an expression has three terms where two are perfect squares and the third is twice their product, it's a hidden $(a \pm b)^2$. Same watch-out applies for cubes (look for 1:3:3:1 coefficients).

TRICK **Plug-in trick for identity MCQs**

When options are large expressions but the question just asks for a numeric value, assign easy numbers such as a = 1, b = 2 to all options. The wrong identities will produce different results and get eliminated in seconds.

Surds and Indices

FORMULA **Multiply same bases**

$$a^m \times a^n = a^{m+n}$$

a is the common base, m and n are any real exponents.

Example: $2^3 \times 2^4 = 2^7 = 128$.

FORMULA **Divide same bases**

$$\frac{a^m}{a^n} = a^{m-n}$$

Subtract exponents when dividing powers of the same base.

Example: $5^6 / 5^2 = 5^4 = 625$.

FORMULA **Power raised to power**

$$(a^m)^n = a^{mn}$$

Multiply the two exponents.

Example: $(3^2)^5 = 3^{10}$.

FORMULA **Negative exponent**

$$a^{-n} = \frac{1}{a^n}$$

Turns a reciprocal into a positive index.

Example: $10^{-3} = 1/10^3 = 0.001$.

FORMULA **Fractional exponent to root**

$$a^{1/n} = \sqrt[n]{a}$$

Links indices to surds.

Example: 27^{1/3} = $\sqrt[3]{27} = 3$.

FORMULA

Product of square roots

$$\sqrt{p} \times \sqrt{q} = \sqrt{pq}$$

Valid when p and q are non-negative.

Example: √2 × √8 = √16 = 4.

TRICK

Rationalising denominator with conjugate

For a denominator like √a + √b, multiply top and bottom by its conjugate √a − √b.

Example: 1/(√5 + √3) → multiply by (√5 − √3)/(√5 − √3) ⇒ (√5 − √3)/(5 − 3) = (√5 − √3)/2.

CONCEPT

Scientific notation snapshot

A number is in scientific form if written as N × 10^k where 1 ≤ N < 10, k is an integer.

Example: 0.00052 = 5.2 × 10^{-4}.

EXAMPLE

Simplifying a nested surd

Simplify √(50) + √(18).

1. Break each number into square factors: 50 = 25·2, 18 = 9·2.
2. √50 = √25·√2 = 5√2; √18 = √9·√2 = 3√2.
3. Add: 5√2 + 3√2 = 8√2.

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